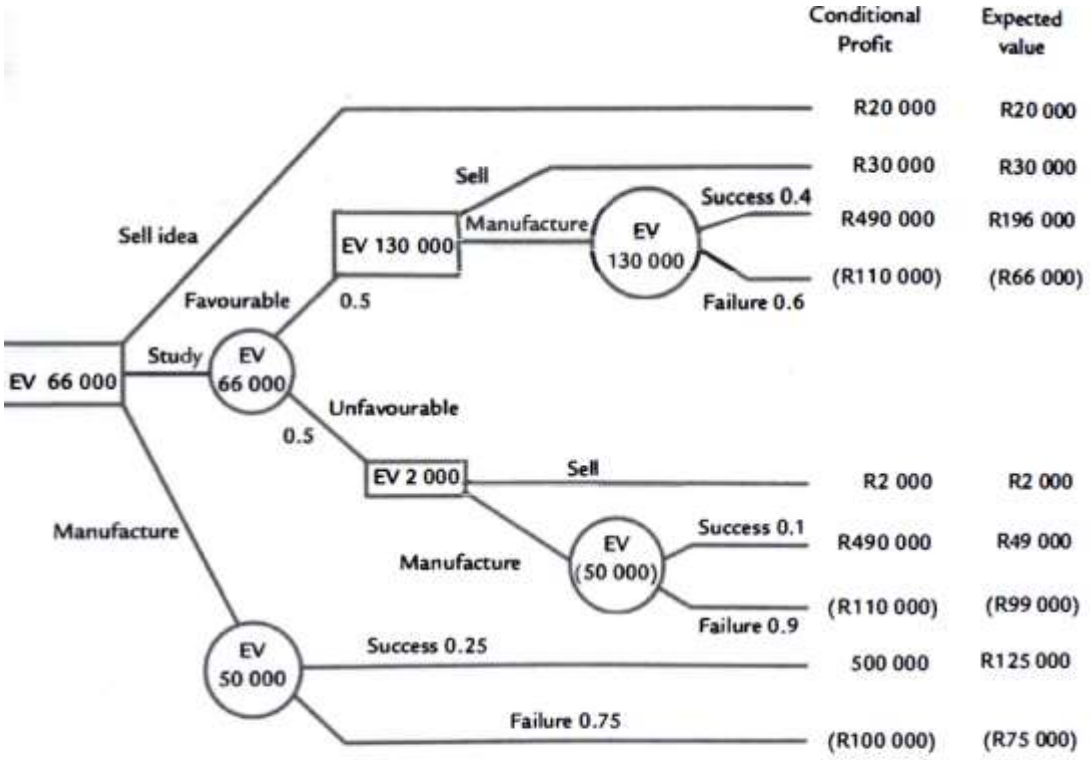
Class exercise – Risk and uncertainty

HFSC Ltd has developed a promising new product. The business has three options available. It can sell the idea of the new product to a company for R20 000, it can hire a consultant to study the market and then make a decision, or it can arrange financing for building a factory and then manufacture and market the product. The study will cost HFSC Ltd R10 000, and its management believes that there is a 50-50 chance that a favourable market will be found. If the study is unfavourable, management believes that it can still sell the idea for R12 000. If the study is favourable, it believes that it can sell the idea for R40 000. But even if a favourable market is found, the chance of an ultimately successful product is about 40%. A successful product will earn a profit of R500 000. Even with an unfavourable study, there is only a 10% chance that a successful product can be expected. If HFSC Ltd's management decides to manufacture the product without a study, there is only a 25% chance of being successful. A product failure costs R100 000.

Required:
Draw a decision tree and determine the course of action that would maximise expected profits for HFSC Ltd.

Solution to class activity



The initial decision would be to conduct the study, as it yields the greatest expected value of a profit of R66 000. The subsequent decisions would be, if the study is favourable, to manufacture the product; and if the study is unfavourable, to sell the product.